

SRM VEC
DEPARTMENT OF AIDS

Continuous Assessment Test - 1

(Regulations 2023)

AD3445 – AI1

Branch: B.Tech | Date: 01.02.2026 | Max. Marks: 100 | Duration: 3 Hrs

Answer ALL Questions

(100 Marks)

PART A — (10 × 2 = 20 marks)

1. Define Artificial Intelligence according to the 'Acting Rationally' approach. [CO1] [BTL1-Remember] (2)
2. Specify the PEAS description for an Automated Medical Diagnosis System. [CO1] [BTL3-Apply] (2)
3. Differentiate between a 'State' and a 'Node' in search terminology. [CO2] [BTL2-Understand] (2)
4. When is a search algorithm said to be 'Optimal'? [CO2] [BTL1-Remember] (2)
5. State the condition for a heuristic function to be 'Admissible'. [CO3] [BTL1-Remember] (2)
6. Identify the main drawback of Greedy Best-First Search. [CO3] [BTL2-Understand] (2)
7. Define a 'Zero-Sum' game in the context of Adversarial Search. [CO4] [BTL1-Remember] (2)
8. What is the purpose of the Alpha and Beta values in Alpha-Beta pruning? [CO4] [BTL2-Understand] (2)
9. Define Constraint Satisfaction Problem (CSP) with its three components. [CO5] [BTL1-Remember] (2)
10. Explain the concept of 'Arc Consistency' in CSP. [CO5] [BTL2-Understand] (2)

PART B — (5 × 16 = 80 marks)

11(a) (i) Explain the different types of Intelligent Agents with neat diagrams. (ii) Describe the various types of environments in AI with examples.

OR

(b) Elaborate on the 'Turing Test' for AI. Discuss the capabilities a computer would need to pass the test and the arguments against it.

12(a) (i) Compare Breadth-First Search and Depth-First Search based on Time, Space, Completeness, and Optimality. (ii) Illustrate the Uniform Cost Search (UCS) algorithm with a suitable example.

OR

(b) Explain the Iterative Deepening Depth-First Search (IDDFS) algorithm. Prove why it is preferred for large search spaces with unknown depth.

13(a) Explain the A* Search algorithm. Demonstrate its working on an 8-puzzle problem or a map-routing problem, showing the $f(n) = g(n) + h(n)$ calculations at each step.

OR

(b) (i) Discuss the Hill Climbing search algorithm and its limitations: Local Maxima, Ridges, and Plateaux. (ii) Explain the concept of Simulated Annealing.

14(a) Explain the Minimax algorithm in game playing. Trace the Minimax decision for a game tree of depth 3 with a branching factor of 2, using arbitrary leaf values.

OR

(b) Apply Alpha-Beta Pruning on a specific game tree (provide a tree with at least 12 leaves). Show which branches are pruned and explain the pruning condition (ϵ & R^2).

15(a) Solve the following Cryptarithmic problem: SEND + MORE = MONEY. Explain the constraints and the backtracking steps involved.

OR

(b) (i) Explain the Backtracking Search algorithm for CSPs and how heuristics like MRV (Minimum Remaining Values) improve it. (ii) Describe the Forward Checking and AC-3 algorithms for constraint

propagation.

Faculty In-Charge

Verified by

HOD